

EKO SAPUTRA

UNITY DEVELOPER PORTFOLIO

PROJECT SHOWCASE

VR STUDY: WORD ARCHITECT

VIRTUAL REALITY (XR)

VR Study: Word Architect is an interactive Virtual Reality educational therapy platform specifically designed for children with dyslexia, transforming traditional learning methods into an immersive 3D environment. Through spatial object manipulation mechanics, players are challenged to assemble randomized letter blocks using Grab & Snap logic to solve word puzzles, which are technically integrated with a state-based progression system to unlock doors and rewards. Supported by an interactive NPC guide and structured dialogue panels across multiple stages, the application effectively enhances focus and letter recognition through a responsive and safe "play-to-learn" VR experience.



Unity Engine

C#

XR Interaction Toolkit

OpenXR

Technical Highlight: Implementation of "State-Based NPC Logic" for responsive stage guidance and a "Key-and-Door" system integrated with 3D block puzzle completion.

```
if (dir != Vector3.zero)
{
    Quaternion targetRot = Quaternion.LookRotation(dir);
    transform.rotation = Quaternion.RotateTowards(
        transform.rotation,
        targetRot,
        rotateSpeed * 100f * Time.deltaTime
    );
}
```

```
void StopMoving()
{
    isMoving = false;
    if (animator != null)
        animator.SetBool("isWalking", false);
    AdvanceStage();
}
```

I developed a custom navigation system for NPCs supporting up to 12 distinct stages. This system ensures seamless synchronization between physical character movement, animations, and state-management progression within the VR environment. By utilizing vector calculations and Quaternions, I achieved smooth, immersive movement and rotation

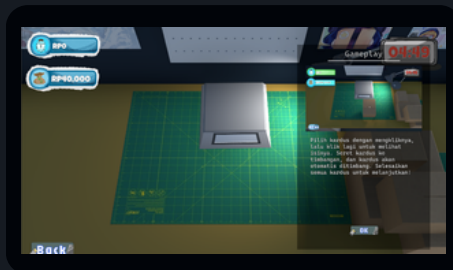
Link Video Gameplay: <https://drive.google.com/file/d/1JEllbScpTHoCpE5Y37TCHucXOqmr2jEx/view?usp=sharing>

Link GitHub Project: https://github.com/mutsaqoful340/VR_Word

E-WASTE MANAGEMENT

DESKTOP INTERACTIVE

E-Waste Management: Interactive 3D Disassembly is an educational desktop simulation designed to raise awareness of electronic waste management through detailed device disassembly mechanics. The technical focus of this project is the development of a precision Drag-and-Drop system and complex 3D Object Hierarchy management, allowing users to detach internal smartphone components—such as the battery, PCB, and screen—for sorting into correct waste categories. Featuring a responsive UI and accurate sorting logic, this application provides a practical technical learning experience in understanding hardware structure and the environmental impact of e-waste.



3D Hierarchy Management

C#

Drag-and-Drop System

OpenXR

Complex 3D object hierarchy management to enable precise disassembly actions and a robust drag-and-drop system for component sorting.

```
void OnMouseDown()
{
    if (isDragging)
    {
        Vector3 mousePos = Input.mousePosition;
        mousePos.z = zOffset;
        Vector3 worldPos = Camera.main.ScreenToWorldPoint(mousePos);
        transform.position = worldPos;
    }
}
```

```
IEnumerator BalikKeAwal()
{
    float t = 0;
    Vector3 start = transform.position;
    while (t < 1)
    {
        t += Time.deltaTime * 3f;
        transform.position = Vector3.Lerp(start, spawnPosition, t);
        yield return null;
    }
}
```

I developed a custom 3D drag-and-drop system for the E-Waste simulation. Key technical implementations include precise screen-to-world coordinate mapping and a Coroutine-based Lerp feedback mechanism that automatically returns objects to their original position if categorized incorrectly. This ensures a smooth, intuitive, and professional user experience.

Link Video Gameplay: <https://drive.google.com/file/d/1CwVOC2vMwUSQq-nlykqueCI6nQxjZLi8/view?usp=sharing>

Link GitHub Project: https://github.com/ekosaputra12/EWasteGame_4

SPRING TIDE

PC / ACTION-ADVENTURE

Spring Tide is a maritime navigation action-adventure game focusing on realistic vessel movement across dynamic water surfaces. The primary technical challenge involved developing a Custom Boat Controller utilizing mass and drag calculations for responsive acceleration, alongside a physics-based Wave Buoyancy system using displacement logic to ensure natural interaction with ocean swells. Additionally, the project features Enemy AI (Kraken/Octopus) implemented with a simplified behavior tree for pursuit mechanics, creating a challenging and immersive maritime experience.



Unity Physics

Buoyancy Systems

C# Scripting

Technical Highlight: Development of a Custom Boat Controller featuring variable acceleration and a responsive steering system driven by real-time wave displacement calculations.

```
if (Mathf.Abs(rudderAngle) > 1f)
{
    float normalizedSpeed = Mathf.Clamp(Mathf.Abs(currentSpeed) / (telegraphStepSpeed * maxRudderAngle));
    float rudderDeflection = Mathf.Abs(rudderAngle) / maxRudderAngle;
    float telegraphFactor = Mathf.Clamp((float)Mathf.Abs(currentTelegraph) / maxRudderAngle);
    float dragStrength = rudderDeflection * normalizedSpeed * telegraphFactor * telegraphStepSpeed;

    if (currentTelegraph > 0)
        targetSpeed -= dragStrength * forwardTurnDrag;
    else if (currentTelegraph < 0)
        targetSpeed += dragStrength * reversalTurnDrag;
}
```

```
if (Physics.Raycast(ray, out RaycastHit hit) && hit.collider.CompareTag("SteeringWheel"))
{
    isDragging = true;

    Vector3 wheelScreenPos = Camera.main.WorldToScreenPoint(steeringWheel.transform.position);
    Vector2 dir = (Vector2)(Input.mousePosition - wheelScreenPos);
    lastMouseAngle = Mathf.Atan2(dir.y, dir.x) * Mathf.Rad2Deg;
}
```

I engineered a complex ship control system featuring a custom Engine Telegraph and real-time synchronized physical instruments. Key technical implementations include a drag-physics-based navigation system where vessel speed is dynamically affected by rudder deflection angles, along with the visual integration of instrument needles and steering wheel rotation using Quaternion and Lerp interpolation for a polished and immersive result.

Link ITCH Project: <https://achronas.itch.io/spring-tide>